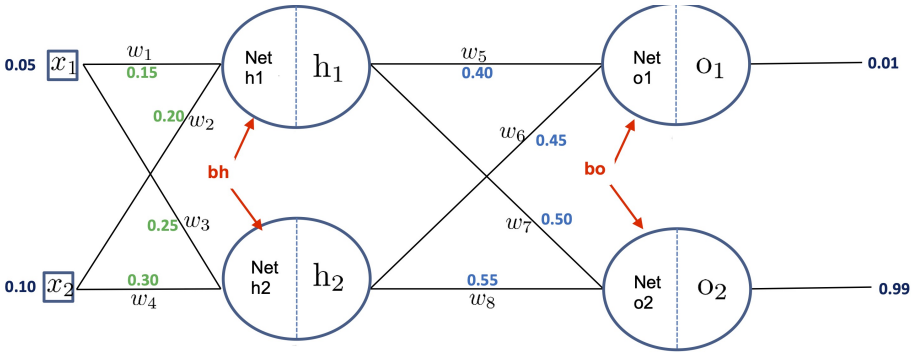
Forward and Backward Pass in Neural Network

September 3, 2025



Assumptions: Inputs: $x_1 = 0.05, x_2 = 0.10$**, Targets:** $t_1 = 0.01,\ t_2 = 0.99$**, Weights (as labeled):** $w_1 = 0.15,\ w_2 = 0.20,\ w_3 = 0.25,\ w_4 = 0.30,\ w_5 = 0.40,\ w_6 = 0.45,\ w_7 = 0.50,\ w_8 = 0.55$**, Biases: hidden** $b_h = 0.35$ **(to both hidden units), output** $b_o = 0.60$ **(to both outputs), Activation:** $\sigma(z) = \frac{1}{1+e^{-z}}$**, Loss:** $E = \frac{1}{2}\sum_k (t_k - o_k)^2$**, Learning rate:** $\eta = 0.50$**.**

1 Forward pass

Hidden layer:

$$\textbf{net}_{h1} = x_1 w_1 + x_2 w_2 + b_h = 0.3775 \ \Rightarrow \ h_1 = \sigma(0.3775) = 0.5932699921$$
$$\textbf{net}_{h2} = x_1 w_3 + x_2 w_4 + b_h = 0.3925 \ \Rightarrow \ h_2 = \sigma(0.3925) = 0.5968843783$$

Output layer:

$$\textbf{net}_{o1} = h_1 w_5 + h_2 w_6 + b_o = 1.1059059671 \ \Rightarrow \ o_1 = \sigma(\cdot) = 0.7513650696$$
$$\textbf{net}_{o2} = h_1 w_7 + h_2 w_8 + b_o = 1.2249214041 \ \Rightarrow \ o_2 = \sigma(\cdot) = 0.7729284653$$

Loss:

$$E = \tfrac{1}{2}[(0.01 - 0.7513651)^2 + (0.99 - 0.7729285)^2] = 0.2983711088$$

2 Backward Pass

Output layer deltas $\sigma'(y) = y(1-y)$.

$$\delta_{o1} = (o_1 - t_1)\, o_1(1-o_1) = 0.1384985616$$
$$\delta_{o2} = (o_2 - t_2)\, o_2(1-o_2) = -0.0380982365$$

Gradients to output weights/biases

$$\frac{\partial E}{\partial w_5} = \delta_{o1}\, h_1 = 0.0821670406, \qquad \frac{\partial E}{\partial w_6} = \delta_{o1}\, h_2 = 0.0826676278,$$
$$\frac{\partial E}{\partial w_7} = \delta_{o2}\, h_1 = -0.0226025405, \qquad \frac{\partial E}{\partial w_8} = \delta_{o2}\, h_2 = -0.0227402422,$$
$$\frac{\partial E}{\partial b_3} = \delta_{o1} = 0.1384985616, \qquad \frac{\partial E}{\partial b_4} = \delta_{o2} = -0.0380982365.$$

Update ($w \leftarrow w - \eta\, \partial E/\partial w$)

$$w_5 = 0.40 - 0.5(0.0821670406) = \mathbf{0.3589164797}$$
$$w_6 = 0.45 - 0.5(0.0826676278) = \mathbf{0.4086661861}$$
$$w_7 = 0.50 - 0.5(-0.0226025405) = \mathbf{0.5113012702}$$
$$w_8 = 0.55 - 0.5(-0.0227402422) = \mathbf{0.5613701211}$$
$$b_3 = 0.60 - 0.5(0.1384985616) = \mathbf{0.5307507192}$$
$$b_4 = 0.60 - 0.5(-0.0380982365) = \mathbf{0.6190491183}$$

Back propagation to hidden layer

$$\delta_{h1} = (w_5\delta_{o1} + w_7\delta_{o2})\, h_1(1-h_1) = \mathbf{0.0087713547}$$
$$\delta_{h2} = (w_6\delta_{o1} + w_8\delta_{o2})\, h_2(1-h_2) = \mathbf{0.0099542547}$$

Gradients to hidden weights/biases

$$\frac{\partial E}{\partial w_1} = \delta_{h1}x_1 = 0.0004385677, \qquad \frac{\partial E}{\partial w_2} = \delta_{h1}x_2 = 0.0008771355,$$
$$\frac{\partial E}{\partial w_3} = \delta_{h2}x_1 = 0.0004977127, \qquad \frac{\partial E}{\partial w_4} = \delta_{h2}x_2 = 0.0009954255,$$
$$\frac{\partial E}{\partial b_1} = \delta_{h1} = 0.0087713547, \qquad \frac{\partial E}{\partial b_2} = \delta_{h2} = 0.0099542547.$$

Update

$$w_1 = 0.15 - 0.5(0.0004385677) = \mathbf{0.1497807161}$$
$$w_2 = 0.20 - 0.5(0.0008771355) = \mathbf{0.1995614323}$$
$$w_3 = 0.25 - 0.5(0.0004977127) = \mathbf{0.2497511436}$$
$$w_4 = 0.30 - 0.5(0.0009954255) = \mathbf{0.2995022873}$$
$$b_1 = 0.35 - 0.5(0.0087713547) = \mathbf{0.3456143227}$$
$$b_2 = 0.35 - 0.5(0.0099542547) = \mathbf{0.3450228726}$$